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Designing for compliance at the microscale: DfAM lessons from a 2PP-printed bellows structure for sensing and actuation

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ABSTRACT: AM enables the design of compliant mechanisms that encode functions directly into geometry. Existing DfAM frameworks rarely address microscale AM, such as two-photon polymerisation (2PP). We present the design process of an airtight, monolithic bellows structure in rigid 2PP resin that serves both as a sensor and an actuator. Through co-evolution of problem and solution, we identify 2PP-specific design considerations and opportunities, including fabrication uncertainties, cross-scale iteration, and design for post-processing, contributing to a case-based DfAM framework for microscale AM.

KEYWORDS: additive manufacturing, two photon polymerisation, design for x (DfX), compliant mechanisms, case study

1. Introduction

With the rise of additive manufacturing (AM), encoding functions such as deliberate compliance into a structure to replace multi-part mechanisms has become more common. Part consolidation is a key topic in design for additive manufacturing (DfAM) (Thompson et al., 2016). Compliant mechanisms, theorized by Howell (2001), rely on bending and elastic deformations in flexure components within the mechanism and are an ever-growing topic that have been accelerated by AM due to the design freedom in three dimensions. The principle of using bending instead of rigid joints have been around for a long time and are used in products like latches, tools and micro-electromechanical systems (MEMS) (Shuib et al., 2007). As components are miniaturized, the traditional approach for assembling functional mechanisms involving sensing, actuation and structure becomes infeasible. This calls the need for designing for function through shape. MEMS already rely on 2.5D compliant structures for sensing and examples can be found in accelerometers, gyroscopes and magnetometers. The edge case of miniaturized AM is the field of two photon polymerisation (2PP). With this technology, we can reach feature sizes down to roughly 100nm and build volumes up to 40x40x40mm for the NanoOne (UpNano). At this scale, multi-component mechanisms become infeasible and embedding function through design becomes a necessity. The design process of such components is not straightforward and require iterative development to achieve desired functionality. Balancing functionality, material choice, feature sizes, and component size require trial and error. Flexures are typically a thinner part of the structure that allows greater deformation with less stress in the material. These are often shorter segments designed to bend. A challenge arises when designing airtight structures, as flexures often introduces openings to control the material stress. On a larger scale, airtight compliant structures such as the bellows structure, Kesling and waterbomb origami demonstrate compliant watertight properties but are often made using multi material methods to reduce the risk of failure in the creases where four sheets meet (Khazaaleh et al., 2022; Liang et al., 2025; Zhao et al., 2021). If made airtight, these structures expand and contract and can be used as both sensors and as an actuator in the presence of pressure difference inside versus outside. Downscaling such a structure to the micro and millimetre

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scale requires a redesign to avoid stress concentrations if kept in a rigid material. Making this structure in a more flexible material can also be done to maximize compliance, but if used as a component in an overall rigid system, this would require assembly. In this article, we present a study on using 2PP and iterative development to achieve a functional combined actuator and sensor using a bellows bladder structure with rigid material together with the design process and lessons learned related to DfAM of such structures.

2. Background and related work

DfAM frameworks enables designers to utilize the design freedom of AM for geometric complexity, function integration and part consolidation. Rosen (2014) and Thompson et al. (2016) have established widely used frameworks for exploring the design space and balancing the AM constraints and opportunities. These frameworks mainly focus on AM for meso- and macro-scale applications and fall short in addressing the growing field of micro- to nano-scale AM. Considerations like feature fidelity, removal of uncured material, post- print handling, and the increased dominance of surface forces overweigh many traditional design considerations. While Buena et al. (2021) provides machine specific guidelines for 2PP and material specific guidelines (van Altena & Accardo, 2023) exists, we lack general 2PP printing design guidelines. Extending DfAM principles to the micro- and nano- scale therefore requires new approaches that addresses these issues.

Compliant mechanisms achieve motion through elastic deformation in regions of the structure called flexures instead of mechanical joints as exemplified in Figure 1a. The key benefits of such a system are the elimination of friction, backlash and assembly, advantages that are critical to miniaturized systems. In MEMS, rigid joints become unfeasible due to surface forces, friction and difficult assembly, while elastic segments move frictionless and responds to force, making compliance often a necessity. Both silicon based MEMS and AM compliant mechanisms are limited by their material yield strain. While silicon is limited to ca 1 % strain before fracture (Khang et al., 2006), polymers often tolerate an order of magnitude more, supporting the potential of AM to expand this field. Power et al. (2018) and Power at al. (2023) successfully demonstrate the fabrication of flexure elements in a complex system for gripping using 2PP printing, and Hu et al. (2025) and Rothermel et al. (2021) demonstrate a compliant spring using 2PP. Typical 2PP photopolymers such as the UpPhoto (UpNano) are relatively stiff which limit deformation if not carefully designed with compliance in mind. More elastic photopolymers exist, but as a monolithic structure this material choice will greatly impact the systems overall rigidity, potentially causing unintended compliance elsewhere in the mechanism. These trade-offs define the design space for miniaturized compliant mechanisms.

2PP printing enables true 3D fabrication with resolution down to the sub micrometres. Localized polymerisation happen in the focal point of the laser and depending on the objective and laser power, this voxel can reach sizes down to 94nm in width for the NanoOne green (UpNano), making it ideal for micro- and nanostructures. The NanoOne (UpNano) supports four objectives: 5x, 10x, 20x and 40x, allowing the user to select a suitable resolution with the compromise of lower volumetric speed at higher resolutions. For each objective, the printer scans within the field of view (FOV) and can build structures larger than the FOV by moving the part relative to the objective and do what is called stitching. Figure 1b illustrates print process and Figure 1c shows an example of a printed part. Examples of the use of 2PP for cases such as robotics and actuators are presented in the review of Rajabasadi et al. (2021). Despite this progress, most 2PP printed devices aiming for compliance rely on either rigid photopolymer exhibiting local deformation in flexures, softer and more flexible materials, or are using 2PP as an intermediate step in combination with a softer material like PDMS exemplified by the Moussi & Kosel (2018) that used 2PP as an intermediate step when creating bellows membranes using 2PP. Achieving airtight and compliant volumes remains difficult using rigid materials and challenges like trapped resin, sealing and stress concentrations create complications. The result is a gap between 2PP printing's potential for compliant mechanisms and the current literature.

The reviewed literature reveal that current DfAM lack guidance for micro-AM. While compliant mechanism theory and 2PP literature each offer partial solutions, designing an airtight, functional monolithic structure with micrometre feature sizes is largely unexplored. To address this gap, we present a case study of designing a 2PP fabricated bellows structure in a rigid photopolymer. This work demonstrates large relative deformation and reflects how function-through-geometry can extend DfAM

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into the microscale. Consistent with Eisenhardt's view (1989) that case studies can serve as a basis for theory development, this case can be used to generalize insights toward a broader microscale DfAM methodology.

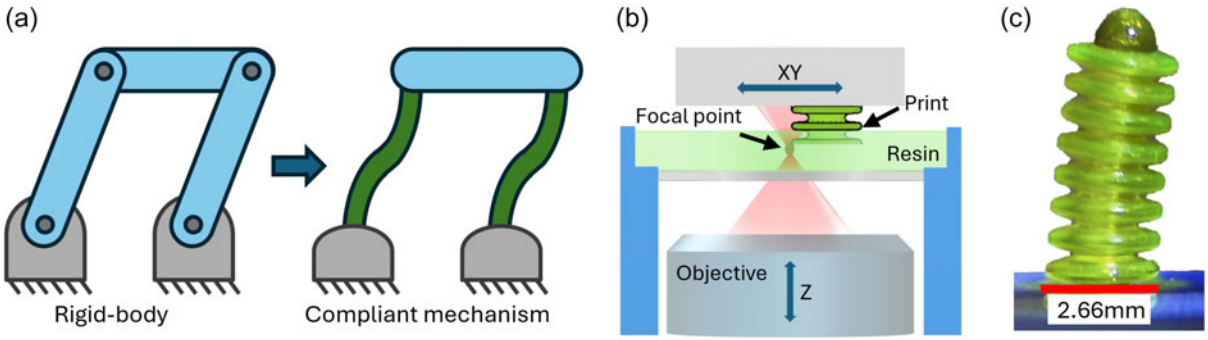


Figure 1. Basic principle of compliant mechanisms where the green part represents the flexures (a), schematic of the 2PP printing process (b) and a 2PP print (c)

3. Methodology and case study

Designing a 2PP printing compatible compliant airtight bladder structure that is actuated by a pressure difference in the internal and external medium has various use cases. Contraction and expansion can be mapped to pressure differences, or inversely, the structure can be used as an actuator responding to applied pressure. If made in a rigid material, this structure can be integrated into a larger mechanism to achieve a more complex task, rather than just expansion or contraction, while keeping the entire mechanism monolithic. This was the initial goal of this case; creating a 2PP printed airtight and compliant structure that reacts to pressure difference for both sensing and actuation through iterative development, using the NanoOne (UpNano) with the UpPhoto material, with the motivation of using it as an actuator for a micro swimmer. In this case, the actuation would be changes in the external pressure outside the bladder, such as pulsating pressure waves in water or inside blood vessels with varying pressure from heartbeats. In this article, we do not go into technical detail but fucus rather on the methodology of this case. **Text in bold** represents both learnings from working with 2PP and methodological aspects used in this case study.

3.1. Iterative development

Dorst & Cross (2001) conceptualize design as a process of co- evolution between problem and solution spaces, where the understanding of the design challenge develops through iterative exploration of potential solutions. Rather than a linear development process, the problem definition emerges through engagement with artifacts and prototypes. This design process reflects the initial state of this study, a vaguely defined problem with an unexplored problem and solution space. The prototyping map is presented in Figure 2.

3.1.1. Early-stage prototyping

To kick off the project with a concept anchor, a 2D micro swimmer demo was made with the intent to verify that an expanding and contracting bladder structure would create asymmetric movement in the fin, similar to how a real fish generates propulsion. This demo serves as a starting point for the iteration cycle as we now concretely **see how the component can actuate a larger system**. Prototypes P1.1 through P1.4 serves to gain understanding of how the compliant mechanism in the 2D concept anchor could be translated into 3D, **switching between micro and macro scale** to gain mechanical intuition for the compliance. The structures investigated was a simple revolved zig-zag pattern designed for single axis compression. The learning outcomes from this initial testing was that the compliance of the structures are highly sensitive to the **relationship between wall thickness and structure size** and an understanding of the overall flexure geometry would be more beneficial rather than optimizing wall thickness.

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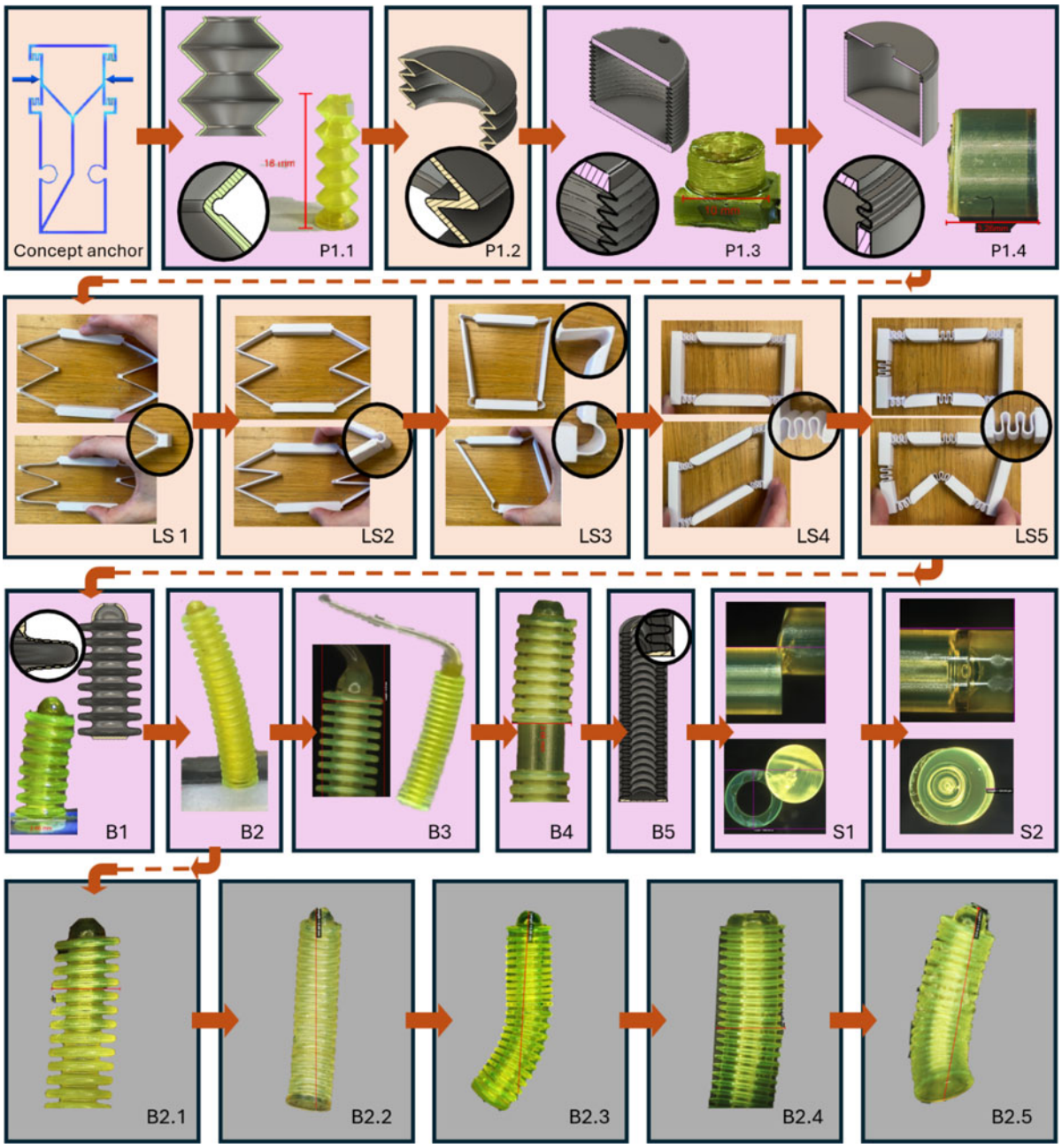


Figure 2. Prototyping map from the iterations in the presented case study colour coded with light orange for large scale exploratory prototypes, pink for 2PP exploratory iterations and grey for refinement prototypes

3.1.2. Macro scale prototyping

When working with microscale compliance, handling, manipulating, and getting a mechanical intuition for the compliance becomes difficult, as one cannot directly see or touch the object in the same was as designers are used to in the meso- and macro- scale. **Iteratively switching from the intended size, and upscaled versions** allow designers to see and feel the design and make intuitive changes before reapplying the design to the microscale. At this point in the prototyping, the specific profile of the "zig-zag" pattern was of interest. Therefore, a series of fused filament fabrication (FFF) printed sections were made (LS1 to LS5) where variations in the corner geometry was investigated. The conclusion was that a smooth snake pattern outperformed the previous zig-zag

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pattern due to the increased length of the flexures giving a more distributed load. This gives the structure less stress concentrations and allows for greater deformation before reaching for the same stress level.

3.1.3. Prototyping of fabrication process

Translating this design into 3D creates up to 90 degrees overhang, making print success uncertain. This uncertainty was quickly eliminated via a simple test print showing **successful 90-degree overhang**, and this marked the first prototype (B1) that show a compliant behaviour and could be compressed without structural failure. At this stage, figuring out the fabrication process to make this structure useful in the context of actuation, meaning sealing and stabilisation was prioritized over finding the optimal relationship between wall thickness and structure size. For the bladder to work as a standalone actuator, it needs to be sealed containing a compressible medium such as air, or alternatively, it could be connected to an external source. When printing using 2PP, the **uncured resin needs to be removed**. This process is often done by dissolving the resin in isopropanol (IPA). This requires some form of **drainage** hole that needs to be sealed post development or connected to a pressure source. In an IPA bath, dissolved **2PP resin tends to flow down** as it is heavier than IPA, so if the drainage hole is in the lowest part of the structure while developing, the internal volume will be replaced with IPA. IPA has a much lower viscosity and will also evaporate, resulting in an easy emptying of the volume. This was already successfully implemented in prototype B1. B1 through B5 was an **exploratory prototyping series**, aimed at stabilizing the structure as it tends to **deform due to its own compliance**. This series included added structural components to the bellows structure to avoid warping. B4 included a cylinder surrounding the bellows structure, and B5 included small pins, both with the intention to stabilize the structure. At this scale, **removing support structures is difficult**, adding extra time-consuming steps and was considered unfeasible. The drainage hole is only a couple of hundreds of microns wide and precise sealing could be difficult, but the structure was made such that the area surrounding the drainage hole served no specific function. A coarser seal could then be applied using hot glue, by dipping the drainage hole into a drop of melted glue. This simple and quick method efficiently sealed the hole, creating an airtight structure. Even though the structure has features in the microscale, creating this larger area for any post processing steps enables **simple sealing without interfering with the functionality**. To avoid design fixation, an alternative sealing method was also explored, using a **two-step printing process**, where the print is developed and inserted back into the 2PP printer to add a final sealing structure onto the print. This alternative method (S1 and S2) proved successful as a proof of concept, but the hot glue process was continued as it was simple in execution, fast and sealed well without interfering with the structure.

3.1.4. Benchmarking and fine tuning of structures

The next challenge was finding the preferred relationship between wall thickness and feature size. To characterize the performance in a realistic but controlled environment, a test rig was constructed using simple mechatronics and a syringe with pressure readings (Figure 3a). This was placed under a microscope to be able to **view the structure while cyclic pressure was applied,** and sync this to the pressure readings (Figure 3b). The movement is exemplified in Figure 3d. This test rig allowed the designer to **verify functionality** in the context of the specific case, and to look for defects during use. This test rig allowed the designers to map pressure to deformation and view potential leakage points. From early prototyping, B2 achieved a reasonable relationship between wall thickness and size, giving clear visible and repeatable deformation without leakage, and was the reference point of the final prototypes. Prototypes B2.1 through B2.5 all explore different wall thicknesses and bellows diameter based on the schematic in Figure 3c using the 5x and 10x objective to explore **trade-offs between compliance, print resolution and size**. Out of these, B2.1, B2.3 and B2.4 achieved visible and repeatable deformations without leakage and concluded this development process. The geometrical parameters and performance of the structures are listed in Table 1. Out of them, B2.4 achieved up to 47.2 % relative change in length.

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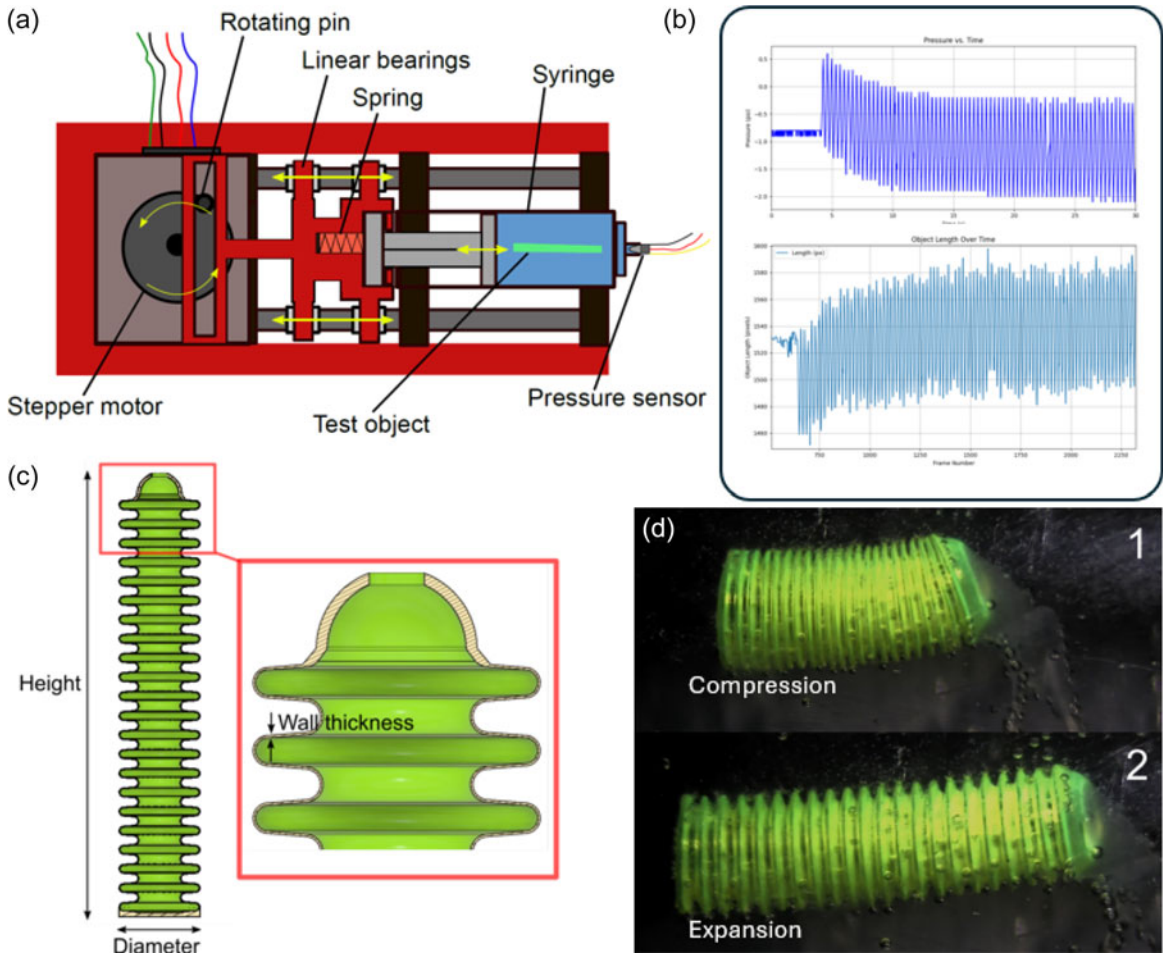


Figure 3. Schematic of the test rig (a), data collected describing pressure and structure length collected for prototyping (b), schematic of final structure (c) and the printed structure during compressing and expansion (d)

Table 1. Geometric parameters and performance of the bellows structure

Prototype	Outer diameter (mm)	Height (mm)	Wall thickness (mm)	Internal bellows radius (mm)	Print mode	Relative length change ΔL (%)	Pressure gradient ΔP (psi)	ΔL/ΔP
B2.1	2.7	14.4	0.023	0.22	5X Fine	26.0	20	1.3
B2.2	1.35	7.2	0.0115	0.11	5X Fine	–	–	–
B2.3	1.35	7.2	0.0115	0.11	10X Fine	1.8	16	0.1125
B2.4	2.7	7.2	0.0115	0.11	10X Fine	47.2	9	5.22
B2.5	0.675	3.6	0.0115	0.055	10X Fine	–	–	–

3.2. Lessons learned

Through the iterative development, lessons learned emerges both in terms of general development for 2PP printing and specifically for compliant mechanisms. These lessons reflect considerations that can inform future developers towards a more general DfAM framework for 2PP at this scale. This section reviews some of the key lessons learned via this case study on compliant mechanisms in the microscale.

3.2.1. Iteration across scales is essential

At the microscale, the designers cannot see, interact and feel the structures directly, which makes iterations hard to evaluate without using specialized techniques. Upscaling prototypes that act as an

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artifact for evaluating new concepts, rather than optimization and refinement, helps overcome this barrier. In this case study, this iteration across scales helped the designers get an intuition for the design before translating the learnings into the microscale. Especially in the early stage of defining the structure, and when choosing the local deformation geometry for the bellows structure.

3.2.2. Designing for post process

Processes such as resin removal, sealing and handling significantly influence the feasibility of 2PP microstructures. Designing with post processing in mind proved essential for this case study. The placement of the drainage holes, the geometry surrounding the drainage holes and the general print orientation of the structure all made post processing easier, shortening the time from print to evaluation. On the other hand, adding support structures like the ones typically found in FFF caused complications after printing. This argues that microscale DfAM extends beyond just the printed geometry itself and needs to include post processing and system integration.

3.2.3. Rapid validation to remove fabrication uncertainties

Overhangs, thin walls, high aspect ratio structures and internal cavities all create printability uncertainties in 2PP printing. Rather than avoiding these uncertainties, small scale prints can quickly validate or dismiss design hypothesis. Similarly, support structures were added but was quickly considered impractical after simple testing. This highlights the value of being agile and using fabrication feedback to inform the design reasoning rather than system level integration and then troubleshooting.

3.2.4. Compliance through geometry rather than mateiral

In 2PP, material choice is limited, and material properties are difficult to alter. Changes in laser intensity and varying degree of polymerisation is doable but requires a deeper understanding of the chemistry involved. Combining multiple materials by multi-step printing involves even more considerations like alignment and the objectives working distance. The deformation and stiffness of the bladder in the case study were almost entirely determined by wall thickness and curvature. This highlights that the already established design principle of compliant mechanisms - function integration through topology rather than material - is translatable into microscale systems.

3.2.5. Design for integration

For the case study presented here, the bladder was used as a standalone sensor/actuator and required only a simple seal to become the final system. The structure included geometries facilitating for this transition between print and system. Had it been used in a system such as the one ideated in the concept anchor, its physical integration into the system needs to be considered, and changes such as drainage hole placement and print orientation needs to be reevaluated to avoid unnecessary assembly steps. This emphasises the need for designing for functional interfacing.

4. Discussion and summary

Many of the lessons learned from this case closely relates to typical DfAM topics and can be transferred into other microscale component designs, also outside of 2PP. Part consolidation (Thompson et al., 2016) is a major topic within DfAM, and the presented structure in this case study consists of two solid ends and a complex multi-story geometry in between, all printed as one part, combining functional compliant elements and rigid ends. Miniaturisation requires part consolidation as assembly becomes infeasible at this scale.

Thompson et al. (2016) also argued that when designing for AM, new opportunities arise together with new constraints. When downscaling and working with 2PP compared to e.g. FFF, new opportunities arise such as 90-degree overhangs, fine feature sizes etc. while limitations like drainage of resin, small part handling, the infeasibility of support structures, and the designers lack of mechanical intuition for microscale parts needs to be considered. For compliant mechanisms, this lack of hands-on feel is especially critical. Being able to bend, twist and press on a prototype normally helps the designer understand how the mechanism moves. At the microscale this is not possible, which is one of the main reasons why iteration across scales becomes so important in this case.

The DfAM worksheet introduced by Booth et al. (2017) covers the design considerations for the meso- to macro-scale of AM. Some elements from this worksheet are transferrable to 2PP printing in the

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micro- and nanoscale, such as stress concentration, complexity and unsupported features. However, its generalisation to cover most AM techniques falls short when designing for 2PP printing. Following the ideas of Eisenhardt (1989), case studies such as the one presented here can later be generalized into a similar format like the design for AM worksheet by Booth et al. (2017).

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